

The Collaborative Ocular Melanoma Study (COMS) Randomized Trial of Pre-enucleation Radiation of Large Choroidal Melanoma: IV. Ten-year Mortality Findings and Prognostic Factors. COMS Report Number 24

COLLABORATIVE OCULAR MELANOMA STUDY GROUP*

- **PURPOSE:** To report rates of death and related outcomes through 10 years after treatment of large choroidal melanoma and to evaluate characteristics of patients and tumors as predictors of relative treatment effectiveness and time to death.
- **DESIGN:** Randomized multicenter clinical trial of pre-enucleation radiation vs enucleation alone conducted as part of the Collaborative Ocular Melanoma Study.
- **METHODS:** Eligible patients were free of metastasis and other cancers at enrollment. All patients were followed for 5 years or longer at scheduled examinations and contacts for metastasis, another cancer, or death. Each decedent was classified as having histopathologically confirmed melanoma metastasis, suspected melanoma metastasis without histopathologic confirmation, another cancer but not melanoma metastasis, or no malignancy.
- **RESULTS:** Within 10 years after enrollment, 576 of 1,003 patients died. Ten-year all-cause mortality rates were 61% for patients in both treatment arms. Ten-year rates of death with histopathologically confirmed melanoma metastasis were 45% in the pre-enucleation radiation arm and 40% in the enucleation alone arm. Older age and larger maximum basal tumor diameter were the

primary predictors of time to death from all causes and death with melanoma metastasis. No differences in unadjusted or adjusted mortality rates were found between treatment arms. Of 448 patients eligible for 10 years of follow-up, 145 patients (32%) were alive and clinically cancer-free 10 years after treatment.

- **CONCLUSIONS:** Longer follow-up confirmed the earlier report of no survival advantage attributable to pre-enucleation radiation. Mortality rates by baseline characteristics should facilitate counseling of patients who have large choroidal melanoma and no evidence of metastasis or another malignancy at diagnosis. (Am J Ophthalmol 2004;138:936–951. © 2004 by Elsevier Inc. All rights reserved.)

UVEAL MELANOMA IS THE MOST COMMON primary intraocular cancer in adults in the United States.¹ The incidence of this cancer remained stable from 1973 through 1997.² Since 1986, the Collaborative Ocular Melanoma Study (COMS) Group has been conducting two multicenter randomized clinical trials of radiotherapy vs enucleation for treatment of choroidal melanoma in patients judged clinically to be free of melanoma metastasis and other cancers at time of diagnosis.³ In the clinical trial which is the subject of this report, patients with choroidal melanoma for whom most ophthalmologists would recommend enucleation because of tumor size or location were assigned randomly to pre-enucleation radiation or to enucleation alone. The first report on all-cause and tumor-related mortality was published in 1998.⁴ As reported at that time, neither a clinically nor statistically significant difference in survival between treatment arms was observed, regardless of whether deaths from all causes or deaths with melanoma metastasis were analyzed. A detailed assessment of status at time of death with respect to melanoma metastasis among the first 435 dece-

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*Prepared by the COMS Steering Committee and other COMS Coordinating Center Investigators on behalf of all members of the COMS group. Lists of members of the COMS Group have been published.^{3,7,9}

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